ANALYSIS REPORT

Track 1 Remaining Family Residual Cellwise Closure Campaign Results Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report closes the residual exact-paper `Track 1` retry wave prepared in:

- `doc/reports/campaign_plans/track1/exact_paper/2026-04-19-01-04-28_track1_remaining_family_residual_cellwise_closure_campaigns_plan_report.md`

  The batch targeted every non-`SVM` family-harmonic pair that still remained 🟡 or 🔴 after the earlier `171`-run cellwise reference wave.
- completed family campaigns: `9`
- completed validation runs: `1026`
- failed validation runs: `0`
- promoted residual pair winners: `90/171`
- aggregate campaign artifact root:

  `output/training_campaigns/track1_remaining_family_residual_cellwise_closure_campaigns_2026_04_19_01_04_28/`

Operational Outcome

- every residual family campaign now exposes:
  - `campaign_leaderboard.yaml`
  - `campaign_best_run.yaml`
  - `campaign_best_run.md`
- the aggregate residual campaign root now exposes the final `1026`-entry

  leaderboard and deterministic bookkeeping representative
- `doc/running/active_training_campaign.yaml` now records the canonical results

  report path for the completed residual retry wave
- the canonical benchmark now reads from the best accepted pair winner across:
  - the earlier `171`-run `cellwise_reference` wave
  - the newer `1026`-run residual retry wave

Family Recovery Outcome

The table below uses compact display aliases for PDF fit. The exact canonical run names remain serialized in the family campaign leaderboards and best-run artifacts under `output/training_campaigns/` .

Family	Amplitude Run	Phase Run	Best Run	Best Scope	Best Closure Score	Open Cells
MLP	A162 a01	P156 a01	A162 a01	amplitudes_only	1.000	0
RF	A0 a03	P156 a01	A0 a03	amplitudes_only	1.000	0
DT	A0 a02	P156 a01	A0 a02	amplitudes_only	1.000	0
ET	A0 a01	P156 a01	A0 a01	amplitudes_only	1.000	0
ERT	A0 a02	P156 a01	A0 a02	amplitudes_only	1.000	0
GBM	A0 a02	P156 a01	A0 a02	amplitudes_only	1.000	0
HGBM	A0 a03	P156 a01	A0 a03	amplitudes_only	1.000	0
XGBM	A0 a02	P156 a01	A0 a02	amplitudes_only	1.000	0
LGBM	A0 a02	P156 a01	A0 a02	amplitudes_only	1.000	0

Aggregate Ranking

The aggregate ranking below also uses compact run aliases for PDF fit. The exact run names remain preserved in the aggregate leaderboard and family-level campaign artifacts.

Rank	Run	Family	Scope	Paper Cell	Met	Near	Open	Closure Score
1	A0 a02	DT	amplitudes_only	2	2	0	0	1.000
2	A0 a05	DT	amplitudes_only	2	2	0	0	1.000
3	A156 a02	DT	amplitudes_only	2	2	0	0	1.000
4	A162 a02	DT	amplitudes_only	2	2	0	0	1.000
5	A1 a02	DT	amplitudes_only	2	2	0	0	1.000
6	A1 a05	DT	amplitudes_only	2	2	0	0	1.000
7	A240 a01	DT	amplitudes_only	2	2	0	0	1.000
8	A240 a02	DT	amplitudes_only	2	2	0	0	1.000
9	A240 a03	DT	amplitudes_only	2	2	0	0	1.000
10	A240 a04	DT	amplitudes_only	2	2	0	0	1.000
11	A240 a05	DT	amplitudes_only	2	2	0	0	1.000
12	A240 a06	DT	amplitudes_only	2	2	0	0	1.000
13	A39 a01	DT	amplitudes_only	2	2	0	0	1.000
14	A39 a02	DT	amplitudes_only	2	2	0	0	1.000
15	A39 a03	DT	amplitudes_only	2	2	0	0	1.000

Final Batch Winner

The aggregate batch bookkeeping representative is `track1_dt_amplitude_0_closure_attempt_02` from family `DT`.

This is a deterministic winner under the per-run closure-first policy across the full `1026`-entry residual batch. It should be read as a bookkeeping representative of the completed campaign, not as the whole scientific verdict of the promoted pairwise residual upgrades.

- winning scope: `amplitudes_only`
- closure score: `1.000`
- met paper cells: `2`
- near paper cells: `0`
- open paper cells: `0`

Scientific Outcome

- the residual retry wave materially improved the accepted benchmark surface without changing the exact-paper family definitions
- amplitude `RMSE` closure rises to `10/10`
- phase `MAE` closure rises to `9/9`
- phase `RMSE` closure rises to `8/9`
- amplitude `MAE` closure rises to `7/10`
- no harmonic remains fully red in Table `6`
- `Track 1` remains canonically open because the harmonic-level family alignment rule is still not fully satisfied

Final Benchmark Refresh Effect

The canonical benchmark after the residual retry closeout is:

- Table `2` amplitude `MAE`: `7/10` harmonics meet or beat the paper target
- Table `3` amplitude `RMSE`: `10/10` harmonics meet or beat the paper target
- Table `4` phase `MAE`: `9/9` harmonics meet or beat the paper target
- Table `5` phase `RMSE`: `8/9` harmonics meet or beat the paper target
- harmonic-level Table `6` closure: `2/10` fully matched, `8/10` partially matched, `0/10` not yet matched The highest-priority still-open harmonics are now:
- `0`
- `3`
- `39`
- `78`
- `81`
- `156`
- `162`
- `240`

Final Interpretation

This batch is operationally complete and materially stronger than the earlier non-`SVM` cellwise reference wave.

The key final conclusion is

- the residual retry wave is fully closed out
- the repository now has the aggressive `1026`-run residual retry evidence on top of the earlier `171` exact-paper reference runs
- the canonical benchmark and Tables `2-5` now read from the best accepted pair winner across both waves
- `Track 1` is no longer numerically blocked by broad red surfaces; the remaining work is concentrated on harmonic-level family alignment in Table `6`